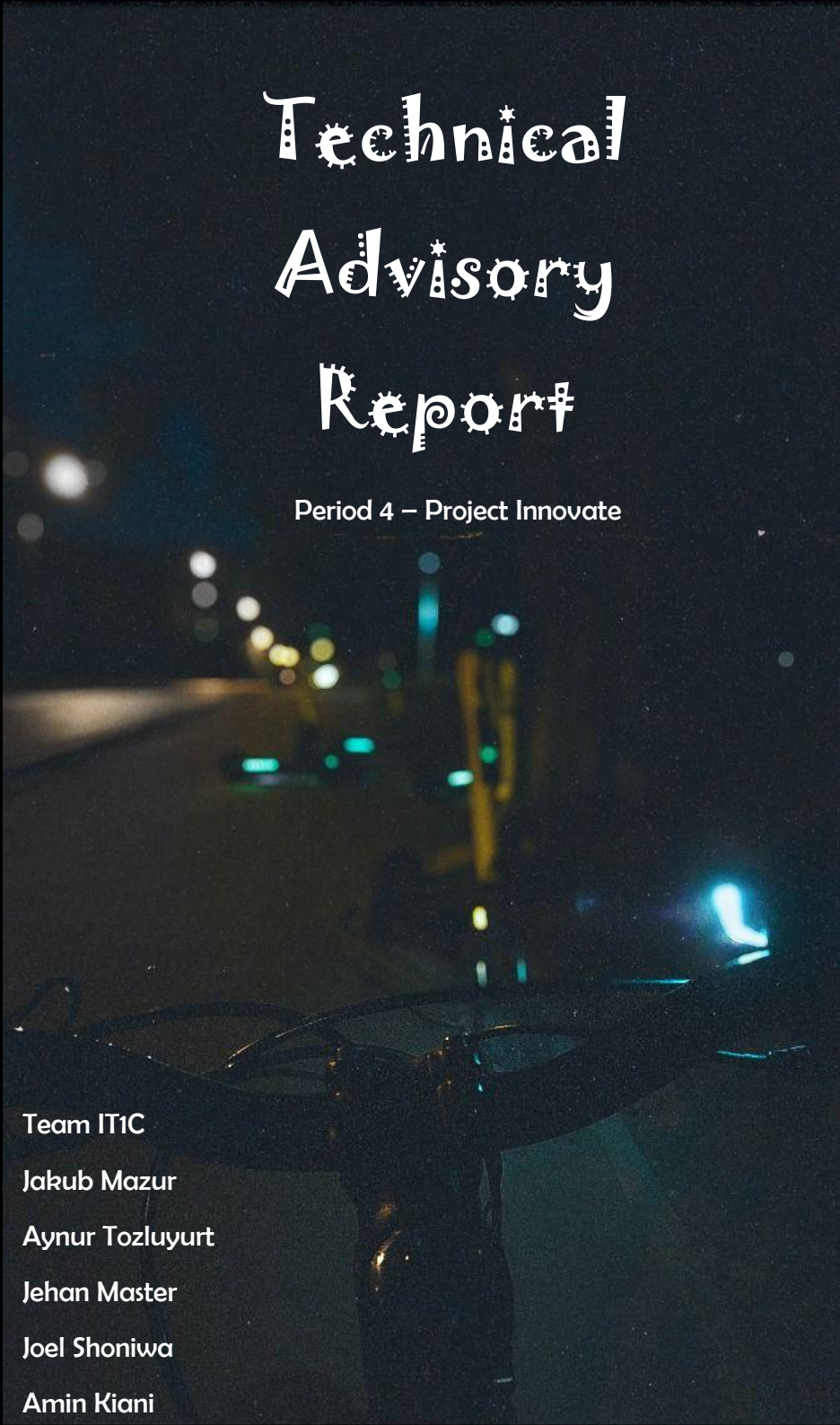




Technical Advisory Report

Period 4 – Project Innovate

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Contents

Summary	4
Introduction.....	5
Structure and Design	6
Ambient Lights and Turning Signal	7
Power Supply and Internal Layout	8
Enclosure and Mounting	8
Hardware Components	9
Software & Control System	13
Recommended Development Environment	14
Main Software Functions	15
System Behaviour Logic.....	15
Sensor Reading and Thresholds	16
Lighting Control.....	16
Servo Control	16
Suggested Code Structure	17
Software Testing Strategy	18
Software Risks and Mitigation.....	19
Software Recommendation.....	19
Project Design Alternatives (Functionalities)	20
– Microcontroller –	20
– Distance Measurement –	21
– Front Light Adjustment –	21
– Power Supply –	22
– Turn Signal –	23
– Night Detection –	23
Implementation Strategy	24
Stage 1 — Sensor & Input Setup	24
Stage 2 — Output Components	24
Stage 3 — System Logic (FSM)	24

Stage 4 — Power & Stability	25
Stage 5 — Final Integration & Testing.....	25
Risks & Considerations	25
Main Project Risks.....	25
Technical Considerations	26
Project Scope Management.....	26
Overall Risk Assessment	26
Conclusion	27

Summary

This Technical Advisory Report explains the design, hardware choices, software structure, and how to build the “Intelligent Bike Light system”.

Our project is designed to improve safety when riding at night. It does this by combining automatic lighting, anti-glare features, and built-in turn signals in one system.

The system uses sensors to measure how bright or dark the environment is. It can turn the front light on automatically when it is dark. It can also reduce brightness when it detects other cyclists or pedestrians to avoid blinding them. Turn signals on the bicycle help the rider show when they want to turn, making their intentions clearer to others.

This report looks at different hardware and software options, compares possible design choices, and explains how the system can be built. It also discusses possible risks during development.

The recommendations in this report are meant to help create a safe, practical, and easy-to-use smart lighting system for bicycles.



Introduction

Cycling is becoming more and more popular for transport and leisure. However, riding in low-light conditions has several safety problems. Cyclists need to be visible to other road users, and they also need to clearly see the road ahead. Normal bicycle lights often use a fixed brightness and angle. This can be uncomfortable or can temporarily blind other cyclists and pedestrians.

The Intelligent Bike Light project was created to solve these problems using sensors, programmable lights, and automatic control. The system can detect when it is dark and turn the bike lights on automatically. It can also change the brightness and direction of the front light when it detects another road user nearby. Additionally, built-in turn signals help cyclists show when they are turning.

The main goal of the project is to improve safety, visibility, and ease of use for cyclists while keeping the system practical and affordable. To do this, the project uses a microcontroller, a distance sensor, a light sensor, LED lights, a servo motor, and user buttons in one system.



Structure and Design

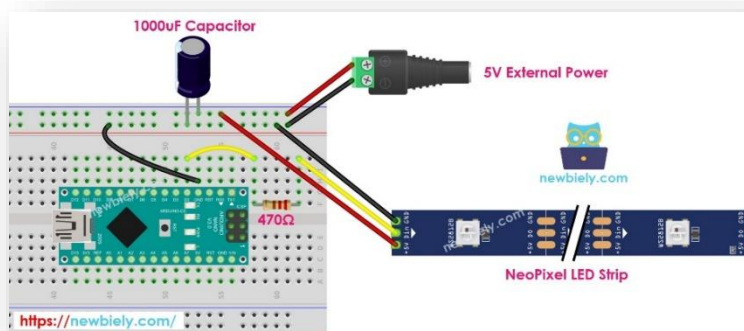
The visual appearance, materials, functional layout and assembly of the intelligent bike lights play a crucial role in ensuring reliability, safety, and seamless integration of all electronic components so that it works up to standard.

Because the device is intended for late night cyclists, its structure must be compact, durable, and capable of withstanding outdoor conditions while maintaining stable performance during motion.

To achieve this, the internal layout should be organized around a 3D printed chassis that protects components such as the microcontroller/sensors/wiring. The enclosure must allow sufficient internal space for the electronic components.



At the same time, the design should prioritize accessibility, enabling quick maintenance or replacement of the battery, wiring, or LED elements without dismantling the entire assembly. Heat shrink tubing, proper cable routing, and a perma proto board will help maintain structural integrity and reduce the risk of accidental disconnections.



Ambient Lights and Turning Signal

A defining feature of the system is the front light which is designed to automatically reduce brightness and tilt downwards when an oncoming cyclist is detected. This mechanism relies on the ToF (Time-of-Flight) distance sensor, which will be positioned at the front of the bike with a clear field of view.

The sensor's data will guide the servo, which adjusts the angle of the bike light. This dynamic movement ensures that the light can be lowered smoothly and quickly, improving safety for both the user and approaching riders.

In addition to the automatic headlight, the design incorporates turn signals to improve rider visibility and communication. Two buttons are mounted on one side of the bicycle handlebar, allowing the cyclist to activate the appropriate indicator by pressing the button.



Power Supply and Internal Layout

The entire system is powered by a lithium polymer battery and will be securely placed to ensure that in cases when the LED is at full brightness the “overheating prospect” is protected and in extreme weather too.

Enclosure and Mounting

The final structure will be housed in a sealed 3D printed chassis designed to withstand outdoor use while maintaining a lightweight profile. The casing should include:

- Rubber band wrapped around the glued gaps to prevent water from entering

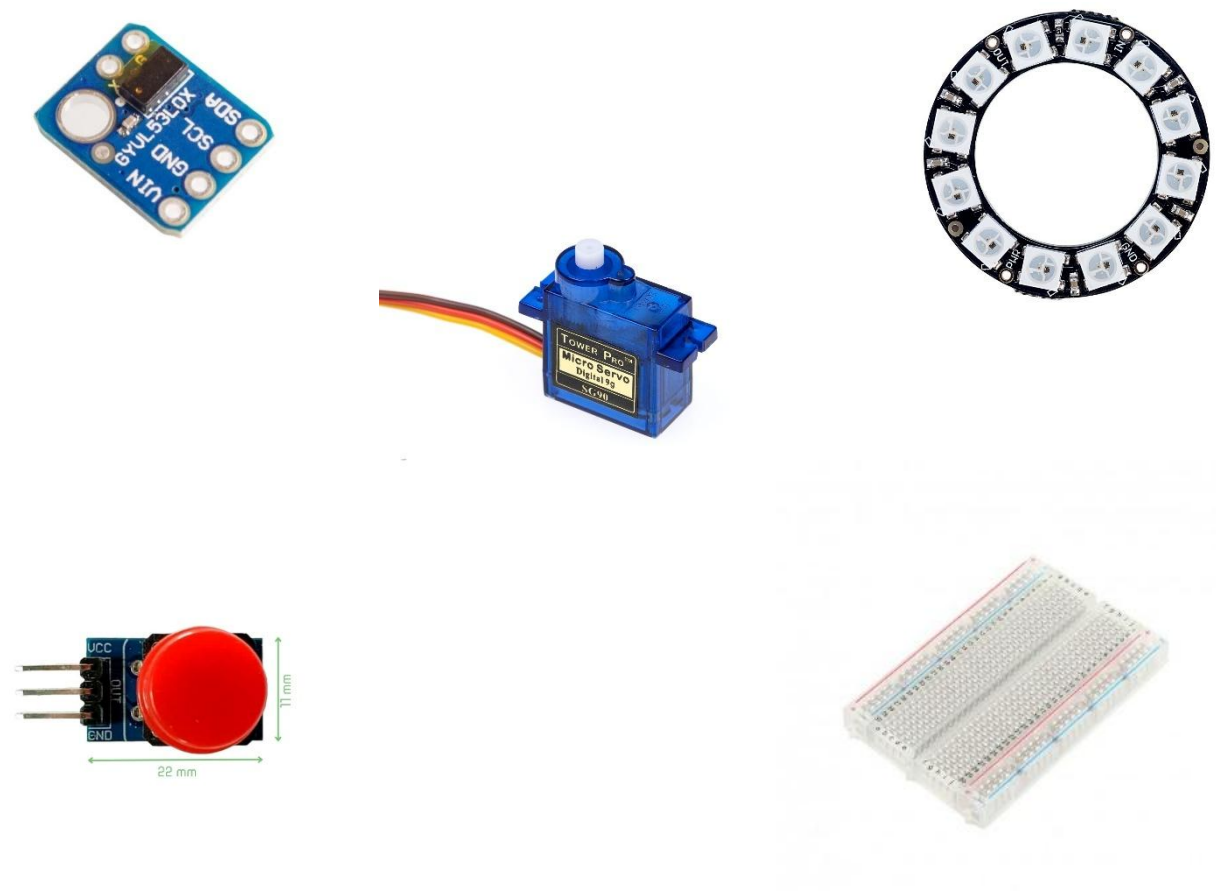
This approach ensures that all components remain protected from accidental impact, moisture, and dust while still allowing the system to be serviced when necessary.

Hardware Components

This section is dedicated to showcasing all the hardware components required for the project prototype and final product. The table below contains components used for the prototype with brief explanations on their functions and choice justification.

Component	Prototype Choice	Function	Reason
Microcontroller	Arduino Nano	Central processing unit that interprets sensor data and executes the control logic required to operate the entire system	Moderate complexity with an output of up to 5V, which simplifies the project goal
Distance Sensor	VL53L0X Laser Distance Sensor	Detection of incoming bikers, cars and pedestrians	Affordable price, using the same ToF (Time-of-Flight) logic as the distance sensor selected for the final product
Light Detector	GL5528 Photoresistor	Night-time detection that activates the entire system	Affordable and delivers satisfactory data
Front Light Movement	MG90S Micro Servo	Front light movement that prevents blinding approaching people	Simple to use and constructed from aluminum and brass to ensure durability
Front Light	NeoPixel Ring - 12 x WS2812 5050 RGBW	Main light source that illuminates the way	Ensures high brightness and features a dedicated white channel

Turn Singal Lights	NeoPixel stick x4 WS2812 RGB (x2)	Display the rider's direction of movement	Ensures high brightness and includes an orange channel for greater visibility
Switches	Push buttons	Used to activate corresponding turn signal light	Simple connection and flexibility
Power Supply	Power Bank 10000mAh	Rechargeable battery that acts as the power supply	Simple connection directly to the microcontroller
Development Board	Breadboard 400 points	Used to temporarily join components to each other	Reliable breadboard that allows flexible component connections



The table below contains recommended components to be used for the final product with brief explanations of their functions and choice justification.

Component	Recommendation	Function	Reason
Microcontroller	Adafruit ESP32-S3 Feather	Central processing unit that interprets sensor data and executes the control logic required to operate the entire system	Integrates powerful processing and built-in battery management
Distance Sensor	TF-mini Plus	Detection of incoming bikers, cars and pedestrians	High physical durability and object detection up to 12 meters
Light Detector	TEMT6000	Night-time detection that activates the entire system	Affordable and reliable
Front Light Movement	MG90S Micro Servo	Front light movement that prevents blinding approaching people	Simple to use and constructed from aluminum and brass to ensure durability
Front Light	NeoPixel ring - 12 x WS2812 5050 RGBW	Main light source that illuminates the way	Ensures high brightness and features a dedicated white channel
Turn Signal Lights	NeoPixel stick x4 WS2812 RGB (x2)	Display the rider's direction of movement	Ensures high brightness and includes an orange channel

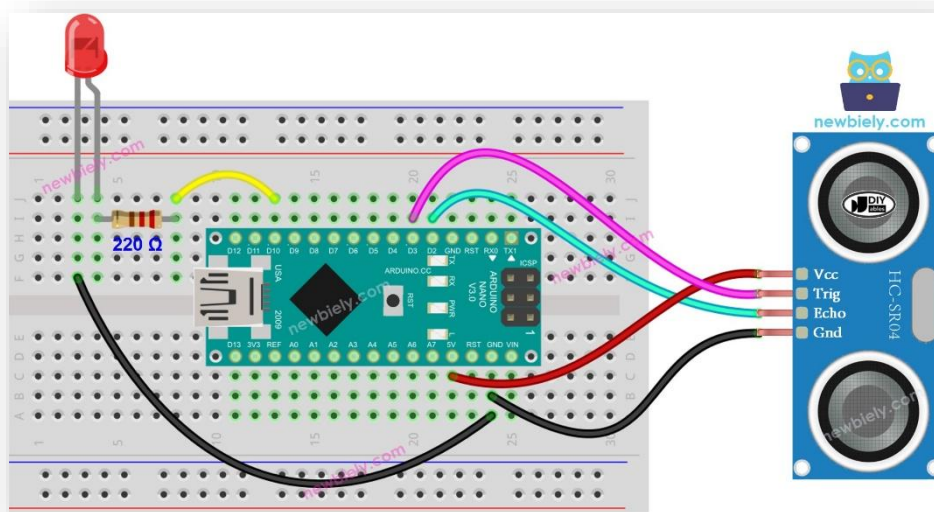
			for greater visibility
Switches	Motorcycle dual handlebar switch	Used to activate corresponding turn signal light	Both switches are placed next to each other, which simplifies mounting and usage
Power Supply	Lithium Polymer Battery 3.7V	Rechargeable battery that acts as the power supply	Reliable battery with sufficient power capacity
Development Board	Adafruit perma-proto half-sized breadboard	Used to permanently join components to each other	Small size and a well-known physical layout
Enclosure	3D-printed enclosure with a rubber seal	Product packaging which protects the components inside	Affordable, lightweight, and easily adjustable depending on component placement

This list only includes the main microcontroller and active modules essential for the project. Standard parts such as wires, resistors, capacitors, and heat-shrink tubing are omitted on purpose to ensure the readability of the table.

Software & Control System

The software of the intelligent bike light is responsible for connecting all electronic parts into one working system. The Arduino Nano will read the values from the light sensor and distance sensor, check the turn signal buttons, control the front NeoPixel ring, blink the left and right NeoPixel sticks, and move the servo motor when the light needs to be lowered. The software should be written in Arduino C/C++ using the Arduino IDE, because this environment is simple, reliable, and suitable for direct control of sensors, LEDs, buttons, and servo motors.

The main software goal is not to create a very complex program, but to create a stable and predictable system. For a safety-related bike light, reliability is more important than unnecessary advanced features. The program should therefore be divided into small functions, so each part can be tested separately before combining everything into the final behaviour.



Recommended Development Environment

The recommended development environment is the Arduino IDE. It supports the Arduino Nano, is easy to install, and allows quick uploading and testing of code. The project can be programmed in Arduino C/C++, which is suitable for real-time hardware control.

Software / Tool	Purpose	Reason for Selection	Risk / Note
Arduino IDE	Writing, compiling, and uploading code to Arduino Nano	Simple and widely used for Arduino projects	Correct board and port must be selected before upload
Arduino C/C++	Main programming language for the controller	Fast and suitable for direct hardware control	Code should be kept clear and modular
Adafruit NeoPixel library	Controls the NeoPixel Ring and NeoPixel Sticks	Makes LED colour and brightness control easier	Brightness should be limited to reduce power use
Servo library	Controls the MG90S servo angle	Built into Arduino IDE and easy to use	Servo movement should be tested slowly to avoid unstable motion
VL53L0X sensor library	Reads distance from the laser distance sensor	Allows accurate distance reading from the sensor	Needs correct wiring and I2C communication

Main Software Functions

The software should be designed around clear functions. This will make the program easier to understand, easier to debug, and easier to explain during assessment. The most important functions are listed below.

- Read the ambient light level from the GL5528 photoresistor.
- Read the distance value from the VL53L0X sensor.
- Control the brightness and colour of the front NeoPixel Ring.
- Move the MG90S servo to raise or lower the light angle.
- Read the left and right push buttons for turn signals.
- Blink the left or right NeoPixel Stick in orange when a button is pressed.
- Apply safe default behaviour when sensor values are unclear or unstable.

System Behaviour Logic

The bike light should follow a simple behaviour flow. First, the program checks whether the environment is dark enough. If it is still bright, the front light can stay off or remain at a very low brightness. If it is dark, the front light turns on automatically. Then the program continuously checks the distance sensor. When an object or incoming cyclist is detected within the chosen range, the front light brightness is reduced and the servo lowers the light angle. When the object is no longer detected, the light returns to its normal brightness and position.

State	Input Condition	Software Action	Output Behaviour
Day mode	Photoresistor detects enough light	Keep main light off or very low	Battery saving mode
Night mode	Photoresistor detects darkness	Turn on front NeoPixel Ring	White front light for visibility
Anti-blinding mode	VL53L0X detects object in front	Dim the light and lower servo angle	Less direct light for incoming cyclist
Normal recovery	Object is no longer detected	Restore brightness and servo angle	Normal front light returns
Left signal	Left button is pressed	Blink left NeoPixel Stick orange	Left turn indication

Right signal	Right button is pressed	Blink right NeoPixel Stick orange	Right turn indication
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Sensor Reading and Thresholds

The software must use threshold values to decide when actions should happen. For the photoresistor, a light threshold should be chosen after testing in different environments. This is important because a classroom, a street with lamps, and a dark outdoor road will all give different values. The final threshold should be based on real sensor readings, not only on assumptions.

For the VL53L0X distance sensor, the program should use a detection distance that is realistic for the prototype. If the distance is too short, the light may react too late. If the distance is too long, it may react to objects that are not actually a problem. A good approach is to start with a safe test value and adjust it after practical testing.

Lighting Control

The NeoPixel Ring should normally be set to white because this gives the best front visibility for cycling. The software should control the brightness level instead of only switching the light on or off. This makes the system more intelligent and allows smooth reaction when another road user is detected.

The turn signal lights should use the NeoPixel Sticks with an orange blinking pattern. The blinking speed should be visible but not too fast. The signal should only blink on the side where the button is pressed. If both buttons are pressed at the same time, the software can either blink both sides as a warning mode or ignore one input depending on the final design choice.

Servo Control

The MG90S servo should be controlled with fixed angle positions. One angle should be used for normal lighting, and another slightly lower angle should be used for anti-blinding mode. The movement should not be too extreme because the cyclist still needs enough light to see the road. A small downward movement is safer and easier to control.

To protect the mechanism, the software should avoid moving the servo continuously back and forth because this can make the system unstable and use more power. A short delay or hysteresis can be added, so the light does not keep moving when the distance reading changes slightly.

Suggested Code Structure

The code should be organised into separate functions. This makes the program easier to explain and prevents one long and confusing loop. A suggested structure is shown below.

`setup()`: initialise sensors, LEDs, buttons, and servo

`loop()`: call the main behaviour functions repeatedly

`readLightLevel()`: read the photoresistor value

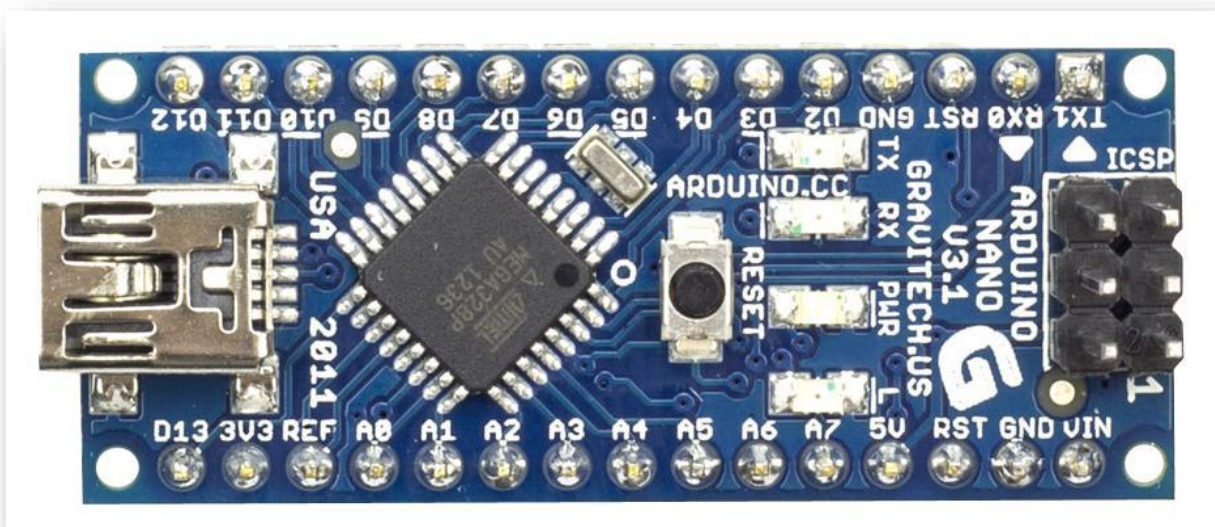
`readDistance()`: read the VL53L0X distance value

`updateFrontLight()`: decide brightness and colour of the front light

`updateServoPosition()`: move the servo only when needed

`updateTurnSignals()`: check buttons and blink left or right signal

`applySafetyDefaults()`: keep the system stable if readings are invalid



Software Testing Strategy

The software should be tested in stages. Each component should be tested alone before the full system is combined. This reduces integration problems and makes it easier to find the cause of errors.

Test Stage	What to Test	Expected Result	Reason
1	Upload simple Arduino test code	Arduino Nano responds correctly	Confirms board and port setup
2	Read photoresistor values	Values change between light and dark	Confirms light detection works
3	Read VL53L0X distance values	Distance changes when object moves	Confirms object detection works
4	Test NeoPixel Ring	White light turns on and dims	Confirms front light control
5	Test NeoPixel Sticks and buttons	Left/right signal blinks correctly	Confirms turn signal input and output
6	Test servo movement	Servo moves between safe angles	Confirms mechanical light adjustment
7	Full system test	All functions work together	Confirms final behaviour is stable

Software Risks and Mitigation

Risk	Possible Impact	Mitigation
Wrong threshold values	Light may turn on too early, too late, or react incorrectly	Calibrate thresholds using real sensor readings
Unstable distance readings	Light may dim or move at the wrong time	Use average readings or a small confirmation delay
Servo moving too often	Power use increases and movement becomes unstable	Use fixed positions and avoid repeated small movements
NeoPixels using too much power	System may reset or become unstable	Limit maximum brightness and test current draw
Button bouncing	Turn signal may blink incorrectly	Use software debounce or stable input logic
Code becoming too complex	Harder debugging and higher chance of failure	Use small functions and test each feature separately

Software Recommendation

The recommended software solution is an Arduino-based control program written in Arduino C/C++. This approach is realistic for the project because all required functions can be handled by the Arduino Nano without adding unnecessary complexity. The software should prioritise stable sensor reading, clear lighting behaviour, simple servo control, and reliable turn signal operation.

Overall, the software should be developed step by step. The most important goal is to make the core safety functions work first: automatic night detection, front light control, anti-blinding dimming, servo lowering, and turn signals. Extra features should only be considered after these core functions are fully tested and reliable.

Project Design Alternatives (Functionalities)

During the development of the intelligent bicycle lighting system, many alternative functionalities and (technical) approaches were discussed and considered before selecting the final design. The purpose of comparing different solutions was to ensure that the final product would provide high reliability, efficiency, and ease of implementation for cycling in the dark.

The team compared multiple design strategies for distance measurement, (front) light adjustment, power supply, turn signal activation, nighttime detection, and mounting methods. Each alternative was carefully evaluated based on factors such as cost, complexity, performance, maintenance, and overall usability.

The following sections present the considered alternatives, their advantages and disadvantages, and the reasoning behind the final design decisions.

– Microcontroller –

The project requires a compact microcontroller capable of handling sensor input, LED control, and servo motor operation.

Option	Advantages	Disadvantages	Complexity	Reliability	Final Decision
ESP32	More processing power	High power consumption, unnecessary extra features	Medium	High	Rejected
Arduino Nano	Compact size, sufficient I/O pins	Lower processing power	Low	High	Selected

The team selected the Arduino Nano because it provides sufficient processing capability for the project while remaining compact, energy efficient, and easy to integrate with the selected sensors and lighting components.

– Distance Measurement –

A major functionality of the project is detecting incoming cyclist or obstacles in order to automatically adjust the front light brightness and angle.

Option	Advantages	Disadvantages	Complexity	Reliability	Final Decision
Ultrasonic Sensor	Cheap, simple implementation	Poor outdoor accuracy, bulky design	Low	Low	Rejected
IR Sensor	Compact and low-cost	Limited range outdoors	Low / Medium	Medium	Rejected
ToF Sensor (TFmini/TF-Luna)	Accurate, fast response time	Higher cost, more advance integration	Medium	High	Rejected
VL53L0X Laser Distance Sensor	Accurate measurements, affordable and suitable for prototyping	Shorter range compared to TF-Luna	Medium	High	Selected

The team initially considered the TF-Luna due to its high accuracy and strong outdoor performance. However, because the project is intended as a prototype, the team ultimately selected the more affordable VL53L0X Laser Distance Sensor. The VL53L0X provides enough functionality for detecting incoming cyclists while reducing overall project cost and implementation complexity.

– Front Light Adjustment –

The intelligent front light requires a mechanism capable of reducing glare for incoming cyclists while maintaining visibility for the rider.

Option	Advantages	Disadvantages	Complexity	Reliability	Final Decision
Brightness Adjustment Only	Simple implementation, fewer components	Glare may still occur	Low	Medium	Rejected
Brightness paired with Servo Movement	Improved glare reduction, more adaptive lightning system	additional mechanical complexity	Medium	High	Selected

and moving
parts

The team selected a combination of brightness adjustment and servo movement because reducing brightness alone may not sufficiently prevent glare. By also lowering the angle of the front light using a servo motor, the system can provide safer and more effective illumination.

– Power Supply –

The project requires a portable and rechargeable power source capable of supplying stable voltage to the microcontroller, LEDs, sensors, and servo motor.

Option	Advantages	Disadvantages	Complexity	Reliability	Final Decision
LiPo Battery	Compact, lightweight, professional embedded-system solution	Requires voltage regulation,	High	High	Rejected
Power Bank	Stable 5V output, rechargeable, safer for prototyping	Larger physical size and less integrated design	Low / Medium	High	Selected

The team selected a USB power bank because it provides a stable 5V power source without requiring additional voltage regulation circuitry. This significantly simplifies development and reduces the risk of power instability during prototyping.

– Turn Signal –

The project includes left and right indicator lights mounted on the steering wheel to improve visibility during turns.

Option	Advantages	Disadvantages	Complexity	Reliability	Final Decision
Automatic Steering Detection	Fully automatic operation	Possibility of false detection	High	Medium	Rejected
Manual Handlebar Buttons	Reliable, simple implementation	Requires user input	Low	High	Selected

The team selected manual handlebar buttons because they provide a more reliable and realistic solution for turn signal activation. Automatic steering detection would introduce additional complexity and may lead to incorrect triggering during normal steering adjustments.

The turn signals are implemented using **two NeoPixel Stick - 4 x WS2812 5050 RGB LED modules** controlled through handlebar-mounted push buttons.

– Night Detection –

The intelligent lighting system should only activate during low-light conditions in order to conserve energy and improve usability.

Option	Advantages	Disadvantages	Complexity	Reliability	Final Decision
Manual	Easy to implement, minimal hardware requirements	User may forget to activate the system	Low	Low	Rejected
Ambient Light Sensor (Photoresistor)	Automatic activation, improved user convenience	Requires calibration under different lighting conditions	Medium	Medium / High	Selected

The team selected the GL5528 Photoresistor because it allows the system to automatically detect nighttime conditions. This improves usability and ensures that the intelligent lighting system operates only when necessary.

Implementation Strategy

To keep the project organized and achievable within the available timeframe, the development of the Intelligent Bike Light is divided into clear stages. Each stage focuses on a specific part of the system, reducing integration problems and making testing easier. Hardware and software progress together, and each stage must be completed before moving on.

Stage 1 — Sensor & Input Setup

- Connect the GL5528 LDR for ambient light detection
- Connect the VL53L0X distance sensor
- Test and record stable sensor values
- Verify push buttons input

Goal: Reliable and consistent input data.

Stage 2 — Output Components

- Install the Neo Pixel front ring
- Install the Neo Pixel turn signal sticks
- Connect and test the MG90S servo
- Calibrate LED brightness and servo angle limits

Goal: All output devices respond correctly to commands.

Stage 3 — System Logic (FSM)

- Implement the three main states:
 - IDLE (Daytime) – Light off, indicators active
 - ACTIVE (Night) – Light on, servo at normal angle
 - OBJECT_DETECTED – Light dimmed, servo dips downward
- Test transitions using real sensor data
- Add fail-safe behavior for sensor errors

Goal: Correct and predictable system behavior.

Stage 4 — Power & Stability

- Add 470 μ F capacitor to stabilize servo power
- Test for brownouts during servo movement
- Move wiring from breadboard to soldered perma-proto board
- Secure all connections with heat-shrink tubing

Goal: Stable, vibration-resistant hardware.

Stage 5 — Final Integration & Testing

- Test the full system on the bike
- Adjust brightness levels and servo angles
- Tune detection thresholds
- Fix remaining software bugs

Goal: Smooth and reliable real-world operation.

Risks & Considerations

Because the project combines electronics, sensors, lighting, and mechanical movement, several risks must be considered to avoid delays and failures.

Main Project Risks

Risk	Description	Impact	Mitigation
Power Drops	Servo draws high current and may reset Arduino	High	Add 470 μ F capacitor on servo rail
Battery Sag	LEDs + servo overload the power bank	Medium	Use a 5V boost converter
Vibration Damage	Loose wires or breadboard failures	High	Use soldered perma-proto board + heat-shrink tubing

Moisture Exposure	Rain may short exposed pins	High	Use sealed enclosure and sensor hoods
Sensor Errors	Distance sensor may misread or fail	Medium	Use Safe State fallback logic

Technical Considerations

Simplicity of Design	Keeps building time low and reduces chances of failure.
Modular Construction	Allows individual parts to be tested and replaced easily.
User Safety	Ensures the rider remains visible, and the system behaves predictably.
Ease of Testing	Supports faster debugging and progress checks.
Expandability	Allows future upgrades such as wireless control or advanced lighting modes.
Battery Accessibility	Simplifies charging and maintenance.

Project Scope Management

To avoid overcomplicating the project, only the essential features are included in the MVP. Advanced features such as wireless control, custom PCBs, or full waterproofing are intentionally left for future development.

Overall Risk Assessment

The overall risk level is manageable as long as the staged development plan is followed. By keeping the scope realistic, testing each subsystem individually, and

prioritizing reliability over unnecessary complexity, the project can be completed successfully within the required timeframe.

Conclusion

The Intelligent Bike Light project provides a practical solution for improving cyclist safety during low-light conditions through automatic night detection, anti-glare lighting adjustment, and integrated turn signals.

The selected hardware and software architecture offer a balance between functionality, cost, and implementation complexity.

While challenges such as power management, sensor reliability, and environmental protection must be addressed, the proposed implementation strategy and risk mitigation measures make the project achievable within the available timeframe.

Overall, the project demonstrates how embedded systems and smart sensing technologies can be applied to create a safer and more user-friendly cycling experience.

